EXPERIMENT - 66

1. Problem Statement

Write a recursive program to **multiply two square matrices**. Do not use iterative loops for the multiplication logic.

2. Algorithm

Main Program

Step 1: START
Step 2: Read n and declare the matrices a, b and c of size n × n.
Step 3: Read the elements of a and b.
Step 4: Call multiply(n, a, b, c, 0, 0, 0).
Step 5: Display the matrix c.
Step 6: STOP

Function multiply(n, a, b, c, i, j, k)

Step 1: If i = n, return.
Step 2: If j = n, call multiply(n, a, b, c, i + 1, 0, 0) and return.
Step 3: If k = n, call multiply(n, a, b, c, i, j + 1, 0) and return.
Step 4: If k = 0, set c[i][j] = 0.
Step 5: Set c[i][j] = c[i][j] + a[i][k] * b[k][j].
Step 6: Call multiply(n, a, b, c, i, j, k + 1).

3. Test Case Table

#	n	Matrix A	Matrix B	Expected Output	Actual Output	Result
1	1	3	4	12	12	PASS
2	2	1 2 / 3 4	5 6 / 7 8	19 22 / 43 50	19 22 / 43 50	PASS
3	3	1 2 3 / 4 5 6 / 7 8 9	9 8 7 / 6 5 4 / 3 2 1	30 24 18 / 84 69 54 / 138 114 90	30 24 18 / 84 69 54 / 138 114 90	PASS

4. Source Code

matmul66.c

```
/* Experiment 66: multiply two square matrices using recursion. */
#include <stdio.h>

void multiply(int n, int a[n][n], int b[n][n], int c[n][n], int i, int j, int k);

int main(void)
{
    int n, i, j;

    printf("Enter the order of the matrices: ");
    scanf("%d", &n);
    int a[n][n], b[n][n], c[n][n];

    printf("Enter the elements of matrix A: ");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    printf("Enter the elements of matrix B: ");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &b[i][j]);

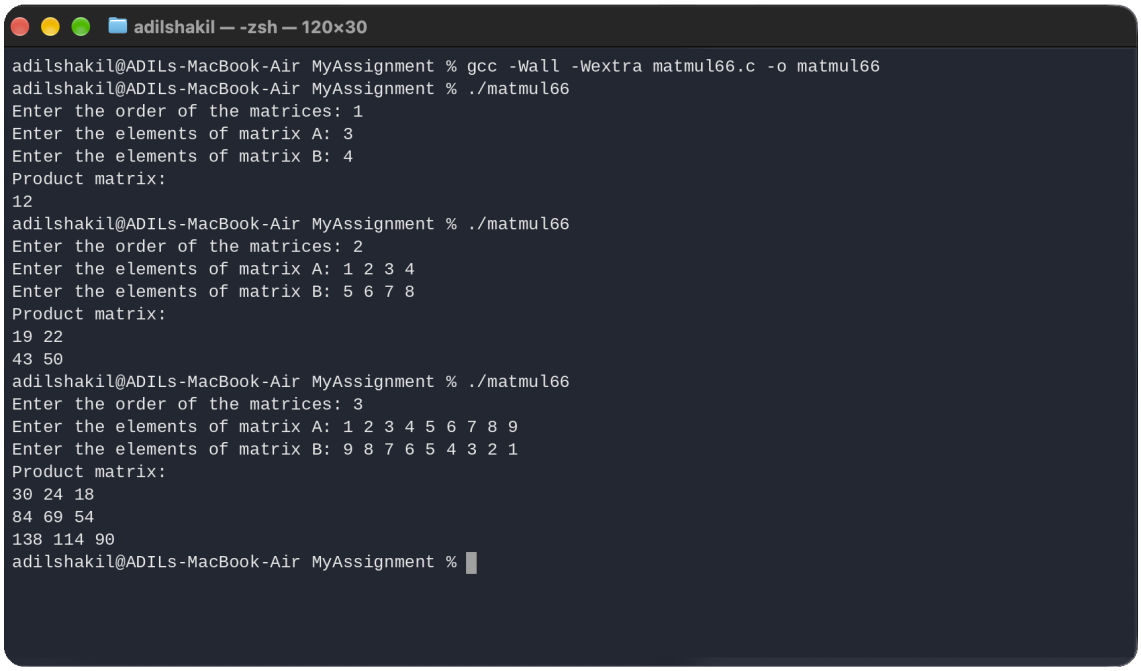
    multiply(n, a, b, c, 0, 0, 0);

    printf("Product matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            printf("%d%c", c[i][j], j == n - 1 ? '\n' : ' ');
    return 0;
}

/* i is the row, j is the column and k walks along them. Each call adds
   one product to c[i][j] and calls itself for the next k. When k reaches
   n it moves on to the next j, and when j reaches n to the next i. */
void multiply(int n, int a[n][n], int b[n][n], int c[n][n], int i, int j, int k)
{
    if (i == n)
        return;
    if (j == n) {
        multiply(n, a, b, c, i + 1, 0, 0);
        return;
    }
    if (k == n) {
        multiply(n, a, b, c, i, j + 1, 0);
        return;
    }
    if (k == 0)
        c[i][j] = 0;
    c[i][j] += a[i][k] * b[k][j];
    multiply(n, a, b, c, i, j, k + 1);
}
```

5. Compilation and Output

Compiled with `gcc -Wall -Wextra matmul66.c -o matmul66` so the small warnings would show up too. It built clean.



All three passed.